Google Skills

What you've accomplished

You've completed the foundational course on AI Agents in just 60 minutes and now possess a clear understanding of what agents are, how they work, and when to use them. This theoretical foundation will serve as your north star as you progress to building real agent systems in upcoming courses.

Your mental models

You now have powerful frameworks for thinking about agents:

1 The three analogies

- **LLMs = consultants** (give advice, don't implement).
- **LLMs with functions = assistants with tools** (can use tools when told).
- **Agents = Trusted employees** (you give them goals, they figure it out).

2 The agent equation

```
Agent = model + tools
        + orchestration
```

Simple but complete.
All three components are essential.

3 The restaurant model

- Model = Cook (knows recipes, makes decisions).
- Tools = Kitchen equipment (actually cooks).
- Orchestration = Process (order, plan, cook, serve).

4 The decision framework

- Multiple steps? Yes.
- Needs adaptation? Yes.
- Requires reasoning? Yes.
- External actions? Yes → **Use an agent.**

What you can do now

With this knowledge, you can:

1. **Differentiate** between LLMs, function calling, and agents with clarity.
2. **Explain** how the three components work together in the agent loop.
3. **Evaluate** whether a problem needs an agent or a simpler solution.
4. **Design** high-level agent architectures for specific use cases.
5. **Avoid** common pitfalls and anti-patterns.

Key takeaways to remember

As you move forward, keep these truths in mind:

1. **Agents ≠ chatbots:** Agents are autonomous systems that can perceive, reason, and act.
2. **Architecture matters:** Model + Tools + Orchestration working together creates the magic.
3. **Tools = superpowers:** Without tools, even the smartest model can't touch the world.
4. **Not everything needs an agent:** Sometimes simple automation is the right answer.
5. **Autonomy is the key:** The defining feature of agents is goal-directed autonomous behavior.

Reflection questions

Solidify your learning by considering:

1. **Personal application:** Think of a repetitive task in your work. Does it fit the agent criteria? (Multi-step? Adaptive? Reasoning-intensive?)
2. **Tool selection:** If building a customer service agent, would you use extensions or functions for order cancellation? Why?
3. **Architecture decision:** Would an e-commerce recommendation system benefit more from a single agent or multiple specialized agents?
4. **Use case evaluation:** Take three tasks you do regularly—which would benefit from agents and which are better automated with simple scripts?

Post questions in the community forum.

Resources

From Google:

Intro to AI Agents

A deep dive of agent definitions, capabilities, use cases, and architectural approaches.

Workflows versus agents

Understand how agentic workflows differ from traditional ones, when to use each, and what they both enable for developers.

The Agent Factory – Episode 1: Agents

Explore what makes a true AI Agent, why frameworks matter, and how to build your own using tools like the Agent Developer Kit.

Conversational versus non-conversational AI Agents

See how agentic AI handles multi-turn interactions, prompt templates, and system instructions, and how to apply single-turn LLMs in today's systems.

From the community:

From gen AI to agentic AI: Everything you need to know – Part 1 and Part 2

A two-part series diving into the capabilities of agentic AI and how it differs from traditional AI.

Final words

You've built a rock-solid theoretical foundation in just one hour. Many developers skip this and jump straight to code without understanding the principles. You now have the mental models that will help you make better decisions as you build agents.

You're no longer someone who's heard about *AI agents*. You're someone who understands what they are, how they work, and when to use them.

The difference between dreaming about agents and building them is taking the next step.

The world of agent development awaits.
Your journey from theory to practice begins now.

Quick reference card

Save this for future reference.

What is an agent?
An autonomous system that achieves goals by observing and acting in its environment.

Three components:

1. Model (reasoning)
2. Tools (doing)
3. Orchestration (process)

Agent loop: Perceive → think → act → check → repeat until done

When to use agents:

- Multi-step workflows
- Adaptation requirements
- Reasoning requirements
- External actions needed

When not to use agents:

- Simple single operations
- Deterministic workflows
- Real-time requirements
- FAQ/simple Q&A

Tool examples:

- Pre-built integrations: Ready-made connections to external services.
- Custom functions: Leverage your existing application code.
- Information retrieval: Access knowledge from databases.